

Cora acoustic OFDM modulation test report

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Executive summary

Every acoustic waveform tested so far uses real-valued OFDM audio with QPSK data carriers, five pilots, a 48 ksample/s sample rate, and a 64-sample cyclic prefix. We have not yet compared QPSK against BPSK, 8-PSK, or QAM over the physical audio path. The experiments called v2, v3, rate-2/3, and uncoded are different channel coding and framing modes around the same QPSK constellation.

The best repeatable physical result is:

- v3 punctured rate-2/3 FEC;
- 256-point FFT;
- 2.25--9.75 kHz carrier band;
- eight 384-byte packets in a continuous superframe;
- channel training refreshed every two packets;
- 3,072 bytes delivered and SHA-256 validated;
- 4,705.8 bit/s mean over three consecutive runs;
- zero packet errors and zero retries in all three runs.

This was 17.7% faster than the matched packet-framed rate-2/3 control at 3,997.6 bit/s. The best validated wide-band result remains v3 rate-1/2 at 1.5--12 kHz and 4,288.1 bit/s. Rate-2/3 failed both wide-band attempts, so its current validation is limited to the standard band.

Common physical layer

Parameter	Value
Audio sample rate	48,000 sample/s
Constellation	QPSK
Default FFT	256 points
Default cyclic prefix	64 samples, 1.333 ms
FFT256 carrier spacing	187.5 Hz
Standard carrier band	2.25--9.75 kHz
Standard FFT256 data carriers	36
Per-symbol tracking	Five pilot carriers
Acquisition	Three known training symbols
Header	Three protected copies with majority vote
Payload integrity	CRC-32 plus end-to-end SHA-256
Test path	Cora Z7-10, USB audio dongle, physical audio loop

All payload rates below are validated application payload, not gross QPSK carrier rate. A PASS means the complete received byte stream matched the source SHA-256. A successful run with retries is marked PASS but is not considered a stable recommended profile.

Coding modes tested

Mode	Body protection	Purpose	Current assessment
v2	Three complete interleaved body copies	Initial robust baseline	Reliable but slow
v3 rate-1/2	K=7, 171/133 convolutional code and Viterbi decoding	Robust coded mode	Validated default
v3 rate-2/3	v3 mother code punctured with repeating 1110 pattern	Higher coded throughput	Best standard-band result; wide band failed
Uncoded	Raw QPSK body with CRC-32	Isolate FEC cost and channel threshold	Fast at 192 bytes; unreliable at 384 bytes

Each mode has a distinct wire version, preventing a receiver configured for one body code from silently accepting another.

Main physical-air comparison

Unless otherwise stated, each row transferred 3,072 application bytes through the physical audio path.

Mode and profile	Packetization	Wall time	Payload rate	Errors / retries	Outcome
v2, FFT256, 2.25--9.75 kHz	8 x 384 B	20.406 s	1,204 bit/s	0 / 0	PASS
v3 rate-1/2, FFT256, 2.25--9.75 kHz, before CPU optimization	8 x 384 B	15.217 s	1,615 bit/s	0 / 0	PASS
v3 rate-1/2, FFT256, 2.25--9.75 kHz, optimized	8 x 384 B	7.381 s	3,329 bit/s	0 / 0	PASS
v3 rate-1/2, FFT256, 1.5--12 kHz, optimized	8 x 384 B	5.731 s	4,288 bit/s	0 / 0	PASS
v3 rate-1/2, FFT256, 1.5--15 kHz	8 x 384 B planned	--	--	Exhausted three attempts on packet 1	FAIL
v3 rate-1/2, FFT512, 2.25--9.75 kHz	8 x 384 B	20.065 s	1,225 bit/s	5 / 5	PASS, unstable
v3 rate-2/3, FFT256, 2.25--9.75 kHz, packet framed	8 x 384 B	6.148 s	3,998 bit/s	0 / 0	PASS
v3 rate-2/3, FFT256, 1.5--12 kHz	8 x 384 B planned	--	--	Exhausted retries twice	FAIL
Uncoded, FFT256, 2.25--9.75 kHz	16 x 192 B	5.978 s	4,111 bit/s	0 / 0	PASS
Uncoded, FFT256, 2.25--9.75 kHz	8 x 384 B	8.496 s	2,893 bit/s	3 / 3	PASS, unstable
v3 rate-2/3 continuous, FFT256, 2.25--9.75 kHz	8 x 384 B	5.223 s mean	4,706 bit/s mean	0 / 0 across 3 runs	PASS

The separate first uncoded 384-byte attempt failed after packet 5 exhausted all retries. The first complete FFT512 attempt failed at packet 7; the successful row above is the repeat and still needed five isolated retries. These failures are part of the assessment even though the retained JSON files record the successful complete transfers.

Continuous-superframe repeatability

The final continuous mode shares quiet guards and acquisition overhead across eight fixed packet slots. It retains a separate header and CRC for each packet, refreshes channel training every two packets, and decodes slots while later audio is still being captured. An invalid slot falls back to an isolated standalone retry.

Run	Wall time	Payload rate	Nominal audio	Errors / retries	SHA-256
1	5.234 s	4,695.7 bit/s	3.960 s	0 / 0	Match
2	5.230 s	4,698.7 bit/s	3.960 s	0 / 0	Match
3	5.203 s	4,723.1 bit/s	3.960 s	0 / 0	Match
Mean	5.223 s	4,705.8 bit/s	3.960 s	0 / 0	Match

The matched packet-framed control took 6.148 seconds at 3,997.6 bit/s with 4.693 seconds of nominal audio. Continuous framing therefore reduced mean wall time by 15.0%, reduced nominal audio by 15.6%, and increased payload throughput by 17.7%.

An earlier version refreshed channel training every four packets. The final per-slot implementation then produced two isolated retries in three trials. Changing the refresh interval to two packets added only 40 ms to an eight-packet waveform and completed three consecutive trials without an error.

Packet-size experiment

The early v2 physical tests established the packet-size tradeoff:

Chunk	Packets	Wall time	Payload rate	Errors / retries
96 B	32	28.154 s	873 bit/s	0 / 0
384 B	8	20.406 s	1,204 bit/s	0 / 0
768 B	4	20.09 s	1,224 bit/s	0 / 0
1,024 B	3	29.09 s	845 bit/s	1 / 1

The 768-byte result gained only 1.6% over 384 bytes, while a failed 1,024-byte packet imposed a large retransmission penalty. The coded modes therefore use 384-byte application chunks. Uncoded mode uses 192-byte chunks because its 384-byte packet error probability erased the airtime benefit.

FFT numerology experiment

FFT	Spacing	Useful symbol	CP	Standard-band data carriers	Gross QPSK rate
256	187.5 Hz	5.333 ms	1.333 ms	36	10.800 kbit/s
512	93.75 Hz	10.667 ms	1.333 ms	76	12.667 kbit/s

FFT512 raises nominal gross rate by 17.3%, but its narrower subcarriers and longer symbols were less tolerant of the real channel. The successful run was 63.2% slower than its clean FFT256 rate-1/2 control and required five retries. FFT256 remains the validated numerology.

Bandwidth experiment

Band	FFT256 data carriers	Tested coding	Result
2.25--9.75 kHz	36	v2, v3 rate-1/2, v3 rate-2/3, uncoded	Validated; rate-2/3 continuous is fastest repeatable result
1.5--12 kHz	52	v3 rate-1/2	PASS at 4,288 bit/s
1.5--12 kHz	52	v3 rate-2/3	FAIL in two attempts
1.5--15 kHz	68	v3 rate-1/2	FAIL after all retries
3--9 kHz	Configuration available	None	Not yet physically benchmarked

Wider programmed bandwidth is beneficial only while per-carrier SNR and the audio hardware response remain usable. The 15 kHz edge and wide-band rate-2/3 results show that nominal carrier count is not sufficient evidence of usable throughput.

Modeled CPU-only OFDM baseline

The separate headless CPU baseline retains QPSK, a 256-point complex FFT, 64-sample CP, pilots, multipath, AWGN, CFO, and Doppler. It is a software throughput and CPU-optimization test, not a different acoustic modulation. All benchmark and real-time checkpoints decoded every expected packet.

The unthrottled benchmark improved from approximately 3,231 bit/s in the first measured implementation to 24,226 bit/s after cached channel taps, batched processing, vectorization, and FFT-based synchronization. The real-time mode improved from approximately 3,056 to 15,299 bit/s while preserving the modeled 48 ksample/s timing. Those numbers must not be compared directly with the physical audio tests because the modeled path excludes ALSA, speaker, microphone, and acoustic retransmissions.

Conclusions and next tests

- 1 Keep QPSK and FFT256 for the current USB audio path.
- 2 Use v3 rate-1/2 when robustness or the 1.5--12 kHz wide profile matters.
- 3 Use v3 rate-2/3 continuous framing for the fastest demonstrated standard band configuration; keep its experimental label until it is exercised over longer runs and changing channels.
- 4 Do not use FFT512, uncoded 384-byte packets, the 15 kHz edge, or wide-band rate-2/3 as defaults.

- 5 The next true modulation comparison should test BPSK versus QPSK under a controlled SNR sweep. Later candidates are differential QPSK and 16-QAM, but 16-QAM should wait until gain stability and soft-decision decoding are available.
- 6 Repeat the selected modes with the intended underwater transducer path. The current report validates the bench audio hardware, not underwater performance.

Machine-readable records

- `results/cora-ofdm-v3-profile-fft256-standard-2026-07-24.json`
- `results/cora-ofdm-v3-profile-fft512-standard-2026-07-24.json`
- `results/cora-ofdm-v3-profile-optimized-2026-07-24.json`
- `results/cora-ofdm-v3-fft256-standard-192-2026-07-24.json`
- `results/cora-ofdm-uncoded-fft256-standard-192-2026-07-24.json`
- `results/cora-ofdm-uncoded-fft256-standard-384-2026-07-24.json`
- `results/cora-ofdm-v3-r23-fft256-standard-384-2026-07-24.json`
- `results/cora-v3-r23-packet-control-standard-air.json`
- `results/cora-v3-r23-continuous-refresh2-run1.json`
- `results/cora-v3-r23-continuous-refresh2-run2.json`
- `results/cora-v3-r23-continuous-refresh2-run3.json`
- `results/cora-ofdm-benchmark.json`
- `results/cora-ofdm-benchmark-tap-cache.json`
- `results/cora-ofdm-benchmark-batch10.json`
- `results/cora-ofdm-benchmark-vector.json`
- `results/cora-ofdm-benchmark-fft-sync.json`

Detailed profiling and implementation analysis is in `docs/cora-ofdm-v3-profile.md`. The original v2 development history and raw packet-size notes are in `docs/cora-ofdm-acoustic-v2.md`.